A Minimal Epistemic Core

Differentiation, Receptive Dependence, and Registration

MARIO ASIS

ABSTRACT. This paper proposes a minimal core of necessary conditions for any epistemic relation: conditions without which nothing could be perceived, registered, or claimed about anything, whatever the domain in which the relation is realized. The core consists of three commitments, none of which can be derived from the others: that the field under consideration admits differentiation; that there is a second relatum admitting differentiation of its own; and that a relation of receptive dependence matches a contrast realized in the field with a contrast in the second relatum, the latter obtaining partly in virtue of the former. Contrasts are throughout between actual terms, never between alternatives, so that the core requires no repertoire of possibilities on either side. On this basis, registration is defined in a strictly structural sense that presupposes neither understanding, nor representation, nor order. The commitments are formal: they presuppose no spatial, temporal, or physical structure, and they are claimed to be necessary, not sufficient. The paper further shows that target and receiver are positions in the relation of dependence rather than natures of its relata, that specifying an articulation selects it without creating it, and that acquisition and retention require an ordering which the core does not contain and whose kind it leaves open.

1. Introduction

Theories of knowledge ordinarily begin with a subject who believes, judges, or is justified, and ask under what conditions such states amount to knowledge. This paper addresses a prior question: what must be admitted for any epistemic relation to be possible at all—for anything to be perceived, registered, or claimed about anything else. Its answer is a set of necessary conditions, not a sufficient account. It identifies what any theory of perception, belief, or knowledge presupposes, and leaves to such theories what makes a relation epistemic in the full sense. The assessment of judgement, in particular, lies outside its scope.

The guiding premise is that an epistemic relation is always a relation to something in particular. To perceive or claim anything about X, X must be

distinguishable from what it is not, and that distinguishing must depend on X; otherwise what is perceived or claimed would not be about X. The commitments introduced below are what this premise requires when no further assumption is made: a differentiation in what is concerned, a differentiation elsewhere, and a dependence of the second on the first.

The conditions are formal. They are stated without reference to space, time, matter, or causation, and they are meant to hold whatever the domain in which an epistemic relation is realized, whether perceptual, physical, linguistic, or mathematical. Examples drawn from particular domains show how the conditions are met there; they do not fix what the conditions are. For the same reason, necessity is claimed only where it can be argued for, and features of the presentation that are conventions rather than commitments are marked as such.

The reconstruction proceeds by minimal commitment. Each condition is introduced only at the point where the account cannot proceed without it, and each is accompanied by a statement of what it does not establish. One methodological caution applies throughout. The exposition necessarily employs ordinary logical resources—identity, distinction, relation, and inference—but using them to state the commitments does not attribute them to what the commitments describe, and in particular not to a knowing subject. A distinction drawn in the exposition is not thereby a distinction drawn by its object.

The proposal has close neighbours. Dretske (1981) analyses the information carried by a signal in terms of nomic dependence and conditional probability, and separates it from meaning and belief. Floridi's diaphoric definition of data takes a lack of uniformity—a difference—as the most elementary constituent of information (Floridi 2011), and his informational structural realism holds that the relation of difference is constitutive of its relata (Floridi 2008). Smith's metaphysics of presence makes *registration* its central notion, as the way in which a world comes to be taken as one of objects (Smith 1996).[1] Enactivist accounts describe cognition as the structural coupling of a system with a medium, both of which belong to a larger organization (Maturana and Varela 1987; Thompson 2007). The present account differs from each in three respects. It separates the differentiability of what is known, the differentiability of what receives it, and the dependence that connects them, and treats each as an independent commitment. It states that dependence between contrasts of actual terms, without probabilities, causes, or alternatives on either side, so that it applies beyond the physical domain and requires no repertoire of possibilities. And it distinguishes the direction of dependence from any order of occurrences, leaving open what kind of order further epistemic operations require.

1. The present use of the term is narrower. Smith's registration is performed by subjects and yields objects; registration here presupposes neither and concerns only the dependence of one contrast on another.

Section 2 introduces the differentiability of the field, and Section 3 distinguishes that capacity from any particular articulation. Section 4 introduces a second relatum, and Section 5 the relation of receptive dependence that connects the two and defines their roles. Section 6 defines registration on that basis and states what it does not include, and Section 7 shows that the resulting structure contains no order. Section 8 summarizes the commitments, the limits of what they establish, what they exclude, and the questions they leave open.

2. Differentiability

Let F designate the *field*: whatever is under consideration. The term fixes the scope of the inquiry; it implies neither a spatial region, nor a collection of discrete objects, nor something already presented to a subject. Nor does the notation establish a boundary intrinsic to reality: F identifies what the account concerns, and any claim about its actual boundaries would require separate justification. The first commitment is the following.

> (D) differentiability
>
> *F admits differentiation, although no particular differentiation is thereby realized or specified.*

Thesis (D) is stronger than the claim that different descriptions of F can be devised. It attributes to F a capacity to enter into a differentiated articulation, so that differentiation is not wholly attributable to the theorist's notation. At the same time, it is weaker than any claim about which articulation is possible: it leaves open whether some articulation is uniquely privileged, and whether its terms would be objects, qualities, regions, values, or something else.

(D) is accordingly a modal commitment; it concerns what F admits. The capacity it attributes belongs to F itself. It is not a stock of possible configurations held anywhere, whether in a mind or in some other repository, and it implies nothing about change or about any order, since the possibility of differentiating something is distinct from the possibility of its changing.

(D) is not deduced from the mere existence of F, for the fact that something is under consideration establishes nothing about the structure it possesses. It is admitted on the following ground. A rival view holds that all differentiation is drawn by whatever receives it: that distinctions are made, not found.[2] The

2. The view has many forms. Spencer-Brown's calculus of indications begins from the injunction to draw a distinction (Spencer-Brown 1969), which Luhmann makes the basis of a constructivist theory of observation (Luhmann 1995); Kant holds that the combination of a manifold can never come to us through the senses (Kant 1998, B129–130), although for him a manifold is still given; and enactivist accounts

present account does not deny that a receiver contributes differentiations of its own; Section 4 requires that it do so. But if every differentiation were supplied by the receiver, nothing in F would correspond to it, and the registration of a difference in F could not be distinguished from the receiver's merely differing within itself. By the guiding premise, what was registered would then not be about F. (D) is the condition under which that distinction can be drawn. It is therefore a necessary condition of the explanation pursued here, not a sufficient account of epistemic possibility. It also involves no knowing subject: that F admits differentiation does not imply that anything differentiates it.

3. Differentiability and Articulation

Differentiability must be distinguished from the realization of any particular differentiation: that F admits differentiation does not entail that a differentiation has been realized.[3]

A particular differentiation of F will be called an *articulation* and written

$$\langle a, \delta, b \rangle_F,$$

where a and b are its *terms* and δ is its *contrast*, the respect in which the terms differ. An articulation is *realized* when F exhibits it, that is, when its terms are actual and differ in the respect δ. The terms of a realized articulation are both actual. They are not alternatives of which one obtains rather than the other, and the contrast is not a choice between them; it is the difference between two terms that are both there. The schema has two terms for simplicity; the account requires only that an articulation have at least two. Nor does the schema presuppose that a and b exist prior to the articulation as independent objects; it marks positions within the articulation described.

Two misreadings of the schema should be excluded. First, δ is not a third item situated between a and b. To treat it as a separator would presuppose spatial organization, whereas δ expresses the differentiation itself. Second, b need not comprise everything that remains once a has been selected. That reading would require an exhaustive partition of F, and a contrast between two terms is not equivalent to a division into "this" and "everything else within the

treat the significance of a difference as brought forth by the system for which it is significant (Thompson 2007).

3. The refusal to ground a difference in terms identified before it has a precedent in Deleuze's critique of the subordination of difference to identity (Deleuze 1994, ch. 1), and his distinction between *differentiation* and *differenciation*—the determination of a structure and its actualization (ch. 4)—is akin to the distinction drawn in this section. Deleuze also rejects the possible as a mere image of the real; the present account shares the refusal of a stock of possibilities, although it retains, in (D), a modal claim about what F admits.

field": the latter adds the requirement that every case within the scope fall on one side or the other. Figure 1 sets the two constructions side by side.

An articulation is *specified* when its terms and contrast are fixed, and it is the account, not F, that specifies it. This might seem to reintroduce the dependence on notation that (D) excludes. It does not. Specifying an articulation selects one of the differentiations that F is claimed to admit; it does not make F admit it, still less realize it. Whether F realizes the specified articulation, and whether anything depends on it, are not settled by the specification, which can therefore fail. Specification fixes what an inquiry concerns, as F fixes its scope; it does not supply what the inquiry finds.

These qualifications mark the first limit of the reconstruction. Differentiability permits articulation, but it neither produces an actual articulation nor determines its form. An account of any particular case must therefore specify the articulation it concerns; it cannot derive that articulation from possibility alone. Moreover, a realized articulation remains an articulation of F, and nothing established so far implies that it is registered anywhere else.

4. The Second Relatum

For an articulation of F to be registered, a differentiation must obtain somewhere other than in the articulation itself. Let R designate a second relatum, distinct from F, in which it obtains.[4] R is not a person, a conscious subject, or an independently bounded system, and it may ultimately prove to be an aspect of a larger organization that also includes F, since a distinction of relata does not entail a distinction of substances.

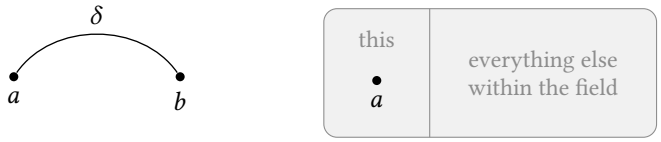
R must admit differentiation of its own. Otherwise the account could describe a contrast in F but nothing in R that would distinguish its registration from its non-registration. The requirement concerns the organization of R, not the possession of concepts or classifications. The second commitment is therefore the following.

(R) second relatum

There is a second relatum R, distinct from F, that admits differentiation of its own.

The terms of a realized differentiation of R will be called its *conditions* and written $r, r', \ldots$. As with articulations of F, the conditions of a realized

4. The vocabulary of reception recalls Kant's characterization of sensibility as receptivity, the capacity to acquire representations through being affected by objects (Kant 1998, A19/B33, cf. A50/B74). The receiver introduced here is thinner: it involves no representations, and nothing is claimed about forms of intuition.



(a) contrast (b) exhaustive partition

FIGURE 1. Two constructions over F. (a) An articulation $\langle a, \delta, b\rangle_F$: a and b mark positions, both actual, and the contrast δ, drawn as a relation rather than as an item between them, is the respect in which they differ. (b) An exhaustive partition, which adds the requirement that every case fall on one side or the other. Only (b) is drawn as a region, because only (b) carries the extensional commitment that a region suggests; in (a) the placement of the marks has no spatial meaning.

differentiation are both actual: r and r' are not alternatives between which R is poised, but distinct conditions that R exhibits. Nothing requires that a differentiation of R have the same form as an articulation of F; the notation names its terms without assigning them any organization.

That (R) is a separate commitment can be shown without a theory of cognition. A contrast in F and a contrast between conditions of R are not the same fact: an articulation of F may be realized while nothing in R answers to it, and R may exhibit contrasts that answer to nothing in F. (R) therefore does not follow from (D); it becomes necessary only when the account passes from differentiation to registration.

(R) also fixes how much individuation the account admits. There must be enough to distinguish F from R, and to take r and r' as conditions of one and the same relatum. There need not be enough to identify enduring objects or successive states of a continuing subject: r and r' are co-present conditions of R, neither alternatives nor stages in its history. To treat these stronger forms of individuation as already established would build into the foundation a persisting, bounded bearer—which is precisely what an account of a knowing subject must eventually explain.

5. Receptive Dependence

Theses (D) and (R) leave F and R unconnected, for a field that admits differentiation and a second relatum that admits differentiation of its own may be entirely unrelated. A further, relational commitment is therefore required. Let C designate *receptive dependence*.

(C) RECEPTIVE DEPENDENCE

Through C, the contrast of a specified articulation realized in F is matched by a contrast between conditions of R, and the latter contrast

obtains partly in virtue of the former.

The thesis has two clauses, and both are needed. The first, *matching*, requires that the terms of the articulation be paired with conditions of R that differ: a with r and b with r', where r and r' are distinct. Were every term paired with the same condition, nothing in R would answer to the contrast δ. Matching alone, however, is not dependence. Two structures may resemble one another without either registering the other,[5] and two contrasts may correspond term by term without either obtaining because of the other (Figure 2). Matching is symmetric: a correspondence that pairs a with r pairs r with a, and can be read from R to F as well as from F to R.

The second clause supplies what matching lacks. The contrast in R obtains *in virtue of* the contrast in F: because of it, and not conversely or merely alongside it. The phrase is not meant to identify C with metaphysical ground; it borrows only two features that the literature on ground has made familiar, namely that dependence of this kind is asymmetric and that it relates what actually obtains, without appeal to what would obtain otherwise (Fine 2012). (C) is accordingly not a counterfactual claim. It does not say what R would be if F were otherwise; it relates two contrasts that are both realized.[6]

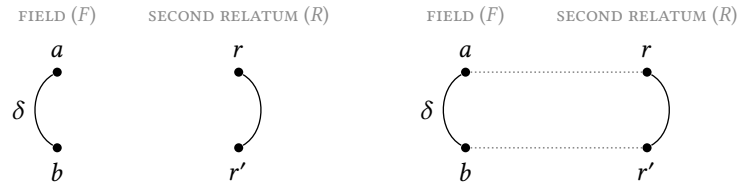
The dependence is partial. Which condition of R answers to which term depends also on the manner in which R is receptive, that is, on the organization of R. The clause requires only that the contrast in R obtain *partly* in virtue of the contrast in F, the remainder being supplied by R itself.[7] The point is captured in Bateson's definition of information as "a difference which makes a difference" (Bateson 1972), anticipated in MacKay's "a distinction which makes a difference" (MacKay 1969), which compresses into one phrase what the present account keeps apart: the first difference belongs to F, the second to R, and the making is the dependence C. For the same reason, the content of (C) should not be concealed behind appeals to "systematic" or "relevant" coupling, which import what the argument has not supplied, namely a standard of relevance or a regularity across cases.

The asymmetry of C defines the roles of its relata. In an instance of C, the *target* is the relatum whose contrast is depended upon, and the *receiver* is the

5. Goodman (1976, ch. 1) makes the parallel point about representation: resemblance is reflexive and symmetric, whereas representation is neither, so resemblance cannot be what representation consists in.

6. Counterfactual analyses of dependence require alternatives to the actual case. Where contrasts are necessary, as in mathematics, they meet a further difficulty: on the standard semantics, counterfactuals with impossible antecedents are all vacuously true (Lewis 1973); for an alternative semantics, see Nolan (1997). Neither difficulty affects (C).

7. This accommodates the enactivist point that a perturbation triggers, but does not specify, the changes it occasions in a system, which are fixed by the system's own structure (Maturana and Varela 1987).



(a) resemblance (b) correspondence

FIGURE 2. Two relations that fall short of receptive dependence. (a) Resemblance: a contrast in F and a contrast between conditions of R share a form, but neither registers the other. (b) Correspondence: a is paired with r, and b with r'. The pairing satisfies the first clause of (C) but is drawn without direction, since it can be read either way. Neither panel establishes (C).

relatum whose contrast depends. Roles are thus positions in a relation, not natures of relata. The same relatum may be target in one instance of C and receiver in another, and two relata may each depend on the other in different respects. This also shows that the account is not circular. Neither (R) nor the definition of C uses the notion of a receiver: C is introduced as an asymmetric relation between contrasts, and the roles are read off it. F and R were named in anticipation of the positions they occupy in the instances considered here.

Two limitations of (C) should be noted. First, the dependence is admitted, not derived from the existence of differentiated relata; a relation cannot be obtained merely by describing its relata. C is introduced because registration requires a connection that neither relatum supplies on its own. Second, the asymmetry of C is primitive in the core. What grounds it in a particular domain is a question for that domain: in physical domains it may be causal, in others it may be logical or constitutive. The core requires that there be such an asymmetry, not that it be of any one kind.

6. REGISTRATION

The three commitments permit a definition of registration in a restricted, structural sense.

DEFINITION OF REGISTRATION

A specified articulation realized in F is registered by R when C holds between them: its contrast is matched by a contrast between conditions of R, which obtains partly in virtue of it.

Registration is thus nothing over and above an instance of C, and it is always registration *of* a specified articulation: to say that R registers F without

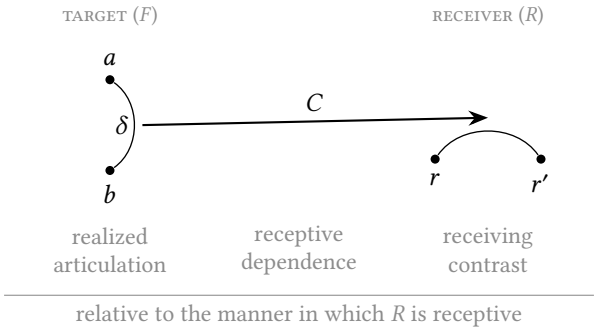


FIGURE 3. Registration. Through C, the contrast δ of the realized articulation $\langle a, \delta, b\rangle_F$ is matched by the contrast between the conditions r and r' of R, which obtains partly in virtue of it. All four terms are actual. The receiving contrast is drawn with a different organization, since it need neither resemble nor reproduce the articulation. The arrow denotes receptive dependence only: it represents neither a transfer nor an order of occurrences, and the columns are positions in C, not regions.

qualification is to leave the claim incomplete. It is also an actual relation between actual contrasts, not a standing disposition of R to respond. Figure 3 depicts the structure so defined.

The definition attributes to the receiver no understanding, identification, or judgement, since none of these capacities has been introduced. Where registration is realized physically, it is found in systems to which nobody would attribute knowledge, and by itself it does not place anything "in the logical space of reasons".[8] Nor does registration move anything from F to R: the receiving contrast need neither resemble the articulation nor reproduce its organization, provided that the dependence obtains. Nor, finally, does it require a representational space, or any repertoire of alternatives, in R. Registration relates contrasts that are realized, and nothing in R need stand for conditions that R does not exhibit.

Registration so defined is supplied by neither relatum in isolation. It comprises three distinct contributions: from the target, the realized articulation; from the receiver, the contrast between its conditions; and between them, the dependence that connects the two. Keeping these contributions distinct prevents registration from being attributed to the target merely because the target is differentiated, or to the receiver merely because its conditions differ. This does not explain the emergence of a knowing subject, but it identifies what such an explanation would have to elaborate.

8. The phrase is from Sellars (1956, § 36); Brandom (1994, ch. 4) develops the contrast between such reliable differential responsiveness and conceptual capacities. In perception, Dretske's non-epistemic seeing, in which an object is visually differentiated from its surroundings without any belief about it, is a case of registration in the present sense (Dretske 1969).

What is registered, on this account, is always a difference within F, never which way F is. The core provides for registering that a differs from b; it does not provide for registering that F is a rather than b, since that would require b to be an alternative to a rather than a term beside it. Accounts of information and knowledge built on that richer form presuppose a set of alternatives among which the actual case is singled out.[9] The core deliberately admits no such set, since it would require a repertoire of possibilities, held in F, in R, or elsewhere, that nothing in the guiding premise demands. Whether registration of which way F is can be reconstructed from contrasts between actual terms, without admitting alternatives, is left open.

7. Registration and Order

The core contains no order of any kind. Its contrasts are not sequences, and the conditions of R are co-present, not successive. C is asymmetric, but the asymmetry of a relation between its positions is not an ordering of occurrences: that the contrast in R obtains in virtue of the contrast in F does not place either before the other.

The core therefore cannot describe the receiver as *acquiring* a contrast, *retaining* it, or *comparing* it with another, and each of these operations requires something different.[10] Comparison requires that two receiving contrasts be available together; this is a relation of co-availability, not of order. Retention requires that one receiver be identified across distinct positions of some ordering. Acquisition requires that a condition of R in which a contrast is unmatched stand in an asymmetric relation to one in which it is matched. Only retention and acquisition require order, and neither follows from (C).

What they require is order in the loosest sense: an asymmetric relation among positions, with no requirement that any two positions be comparable, and with no metric, no linearity, and no universal index. Whether this relation must also be transitive, and so form a strict partial order, may be left open. Temporal succession is one way of realizing such an order; the order of steps in

9. Shannon's theory measures information as selection from a set of possible messages (Shannon 1948), and Dretske's informational content is defined by conditional probabilities over such alternatives (Dretske 1981). In epistemology, relevant-alternatives theories and contrastivism analyse knowledge as the elimination of alternatives (Dretske 1970; Lewis 1996; Schaffer 2004); counterfactual and interventionist analyses of dependence share the same structure (Lewis 1973; Woodward 2003).

10. The three operations closely parallel the synthesis of apprehension, reproduction, and recognition in the first edition of Kant's Transcendental Deduction (Kant 1998, A98–A110), and Husserl's distinction between retention and recollection (Husserl 1991) analyses the second of them. Both treat the operations as presupposing temporal form; the present account requires only order, of which temporal order is one kind.

an inference or a proof is another, which is not temporal. The core is compatible with either and requires neither.

For the same reason, if receptive dependence is written with a directed symbol, such as an arrow from F to R, that symbol should not also serve, without further explanation, to represent order. The asymmetry of dependence and the asymmetry of an ordering play different roles in the account and should be kept notationally distinct.

8. Conclusion

Table 1 summarizes the elements of the account, what each permits, and what each leaves unestablished. The specified articulation is listed alongside the three commitments, although it is specified in each application rather than admitted once and for all.

Table 1. The elements of the account, with what each permits and what each leaves unestablished.

ELEMENT	WHAT IT PERMITS	WHAT IT DOES NOT ESTABLISH
(D) Differentiability	A differentiation attributable to F itself.	Any particular articulation, or its realization.
Specified articulation	Terms and the contrast in which they differ.	Its own realization, registration, exhaustive partition, or objecthood.
(R) Second relatum	Differentiation of R's own.	A knowing subject, or any dependence on F.
(C) Receptive dependence	Registration of a realized articulation; the roles of target and receiver.	Which way F is; understanding, representation, order, acquisition, or retention.

The reconstruction does not derive everything from differentiability. It shows, rather, why further commitments are needed, and it distinguishes those commitments from what they make it possible to define; in particular, neither the second relatum nor its dependence on the target is concealed in the initial claim about F. The resulting proposal is that any epistemic relation, in whatever domain it is realized, requires *differentiation and receptive dependence* between contrasts of actual terms, with target and receiver distinguished as positions within that dependence. The claim is one of necessity only: these conditions are required, by the guiding premise, for anything to be perceived, registered, or claimed about anything, and they are not sufficient for any of these.

The core is minimal, but it is not empty: its commitments exclude several views of what an epistemic relation presupposes. They exclude, first, the view that all differentiation is supplied by the receiver, since registration requires a contrast attributable to the field itself (Section 2). They exclude, second, the view that registration consists in resemblance or in term-by-term correspondence, since both are symmetric and neither supplies the dependence required by the second clause of (C) (Section 5). Third, they exclude any conception of registration as a transfer from target to receiver, since the receiving contrast need neither resemble nor reproduce the articulation (Section 6). Fourth, they exclude the view that a repertoire of alternatives, whether held in the field, in the receiver, or elsewhere, is a precondition of registration, since every contrast in the core is between actual terms (Sections 3 and 6). Fifth, they exclude the view that time, or order of any kind, is such a precondition, since registration has been defined without either (Section 7). Finally, they exclude the view that a knowing subject, concepts, or representations precede registration, since none of these figures in (D), (R), or (C). None of these exclusions denies that the structures concerned are present in particular epistemic relations. What is denied is only that they belong to the core: wherever they are needed, they must enter as further commitments, stated as such.

Four questions remain open. The first is what grounds the asymmetry of C in a given domain. The second is whether receptive dependence can itself be explained in terms of more elementary relations; an arrow whose meaning is left unspecified would not answer it. The third is whether registration of which way F is can be reconstructed without admitting alternatives. The fourth is what order acquisition and retention require, and whether anything stronger than an asymmetric relation is needed. The reconstruction ends at structural registration. Order, memory, recognition, and representation have not been presupposed in reaching it, and none of them has yet been derived from it.

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References

Bateson, G. (1972). *Steps to an Ecology of Mind.* San Francisco: Chandler.

Brandom, R. B. (1994). *Making It Explicit: Reasoning, Representing, and Discursive Commitment.* Cambridge, MA: Harvard University Press.

Deleuze, G. (1994). *Difference and Repetition.* Trans. by P. Patton. New York: Columbia University Press. First published as *Différence et répétition*, Paris: Presses Universitaires de France, 1968.

Dretske, F. I. (1969). *Seeing and Knowing.* Chicago: University of Chicago Press.

Dretske, F. I. (1970). "Epistemic Operators". *Journal of Philosophy* 67.24, pp. 1007–1023.

Dretske, F. I. (1981). *Knowledge and the Flow of Information.* Cambridge, MA: MIT Press.

Fine, K. (2012). "Guide to Ground". In: *Metaphysical Grounding: Understanding the Structure of Reality*. Ed. by F. Correia and B. Schnieder. Cambridge: Cambridge University Press, pp. 37–80.

Floridi, L. (2008). "A Defence of Informational Structural Realism". *Synthese* 161.2, pp. 219–253.

Floridi, L. (2011). *The Philosophy of Information*. Oxford: Oxford University Press.

Goodman, N. (1976). *Languages of Art: An Approach to a Theory of Symbols*. 2nd ed. Indianapolis: Hackett.

Husserl, E. (1991). *On the Phenomenology of the Consciousness of Internal Time (1893–1917)*. Trans. by J. B. Brough. Collected Works 4. Dordrecht: Kluwer.

Kant, I. (1998). *Critique of Pure Reason*. Ed. and trans. by P. Guyer and A. W. Wood. Cambridge: Cambridge University Press. First published 1781 (A) and 1787 (B); cited by the pagination of these editions.

Lewis, D. (1973). *Counterfactuals*. Oxford: Blackwell.

Lewis, D. (1996). "Elusive Knowledge". *Australasian Journal of Philosophy* 74.4, pp. 549–567.

Luhmann, N. (1995). *Social Systems*. Trans. by J. Bednarz Jr. and D. Baecker. Stanford: Stanford University Press. First published as *Soziale Systeme*, Frankfurt am Main: Suhrkamp, 1984.

MacKay, D. M. (1969). *Information, Mechanism and Meaning*. Cambridge, MA: MIT Press.

Maturana, H. R. and F. J. Varela (1987). *The Tree of Knowledge: The Biological Roots of Human Understanding*. Boston: New Science Library.

Nolan, D. (1997). "Impossible Worlds: A Modest Approach". *Notre Dame Journal of Formal Logic* 38.4, pp. 535–572.

Schaffer, J. (2004). "From Contextualism to Contrastivism". *Philosophical Studies* 119.1–2, pp. 73–103.

Sellars, W. (1956). "Empiricism and the Philosophy of Mind". In: *The Foundations of Science and the Concepts of Psychology and Psychoanalysis*. Ed. by H. Feigl and M. Scriven. Minnesota Studies in the Philosophy of Science 1. Minneapolis: University of Minnesota Press, pp. 253–329.

Shannon, C. E. (1948). "A Mathematical Theory of Communication". *Bell System Technical Journal* 27, pp. 379–423, 623–656.

Smith, B. C. (1996). *On the Origin of Objects*. Cambridge, MA: MIT Press.

Spencer-Brown, G. (1969). *Laws of Form*. London: Allen & Unwin.

Thompson, E. (2007). *Mind in Life: Biology, Phenomenology, and the Sciences of Mind*. Cambridge, MA: Belknap Press of Harvard University Press.

Woodward, J. (2003). *Making Things Happen: A Theory of Causal Explanation*. New York: Oxford University Press.